

Chapter 23

Squares Modulo p

Revised Version of Chapter 23

We learned long ago how to solve linear congruences

$$ax \equiv c \pmod{m}$$

(see Chapter 8). It's now time to take the plunge and move on to quadratic equations. We devote the next three chapters to answering the following types of questions:

- Is 3 congruent to the square of some number modulo 7?
- Does the congruence $x^2 \equiv -1 \pmod{13}$ have a solution?
- For which primes p does the congruence $x^2 \equiv 2 \pmod{p}$ have a solution?

We can answer the first two questions right now. To see if 3 is congruent to the square of some number modulo 7, we just square each of the numbers from 0 to 6, reduce modulo 7, and see if any of them are equal to 3. Thus,

$$0^2 \equiv 0 \pmod{7}$$

$$1^2 \equiv 1 \pmod{7}$$

$$2^2 \equiv 4 \pmod{7}$$

$$3^2 \equiv 2 \pmod{7}$$

$$4^2 \equiv 2 \pmod{7}$$

$$5^2 \equiv 4 \pmod{7}$$

$$6^2 \equiv 1 \pmod{7}.$$

So we see that 3 is not congruent to a square modulo 7. In a similar fashion, if we square each number from 0 to 12 and reduce modulo 13, we find that the congruence $x^2 \equiv -1 \pmod{13}$ has two solutions, $x \equiv 5 \pmod{13}$ and $x \equiv 8 \pmod{13}$.¹

As always, we need to look at some data before we can even begin to look for patterns and make conjectures. Here are some tables giving all the squares modulo p for $p = 5, 7, 11$, and 13.

b	b^2
0	0
1	1
2	4
3	4
4	1

Modulo 5

b	b^2
0	0
1	1
2	4
3	2
4	2
5	4
6	1

Modulo 7

b	b^2
0	0
1	1
2	4
3	9
4	5
5	3
6	3
7	5
8	9
9	4
10	1

Modulo 11

b	b^2
0	0
1	1
2	4
3	9
4	3
5	12
6	10
7	10
8	12
9	3
10	9
11	4
12	1

Modulo 13

Many interesting patterns are already apparent from these lists. For example, each number (other than 0) that appears as a square seems to appear exactly twice. Thus, 5 is both 4^2 and 7^2 modulo 11, and 3 is both 4^2 and 9^2 modulo 13. In fact, if we fold each list over in the middle, the same numbers appear as squares on the top and on the bottom.

How can we describe this pattern with a formula? We are saying that the square of the number b and the square of the number $p - b$ are the same modulo p . But

¹For many years during the nineteenth century, mathematicians were uneasy with the idea of the number $\sqrt{-1}$. Its current appellation “imaginary number” still reflects that disquiet. But if you work modulo 13, for example, then there’s nothing mysterious about $\sqrt{-1}$. In fact, 5 and 8 are both square roots of -1 modulo 13.

now that we've expressed our pattern by a formula, it's easy to prove. Thus,

$$(p - b)^2 = p^2 - 2pb + b^2 \equiv b^2 \pmod{p}.$$

So if we want to list all the (nonzero) numbers that are squares modulo p , we only need to compute half of them:

$$1^2 \pmod{p}, \quad 2^2 \pmod{p}, \quad 3^2 \pmod{p}, \dots, \quad \left(\frac{p-1}{2}\right)^2 \pmod{p}.$$

Our goal is to find patterns that can be used to distinguish squares from nonsquares modulo p . Ultimately, we will be led to one of the most beautiful theorems in all number theory, the Law of Quadratic Reciprocity, but first we must perform the mundane task of assigning some names to the numbers we want to study.

A nonzero number that is congruent to a square modulo p is called a *quadratic residue modulo p* . A number that is not congruent to a square modulo p is called a *(quadratic) nonresidue modulo p* . We abbreviate these long expressions by saying that a quadratic residue is a QR and a quadratic nonresidue is an NR. A number that is congruent to 0 modulo p is neither a residue nor a nonresidue.

To illustrate this terminology using the data from our tables, 3 and 12 are QRs modulo 13, while 2 and 5 are NRs modulo 13. Note that 2 and 5 are NRs because they do not appear in the list of squares modulo 13. The full set of QRs modulo 13 is $\{1, 3, 4, 9, 10, 12\}$, and the full set of NRs modulo 13 is $\{2, 5, 6, 7, 8, 11\}$. Similarly, the set of QRs modulo 7 is $\{1, 2, 4\}$ and the set of NRs modulo 7 is $\{3, 5, 6\}$.

Notice that there are 6 quadratic residues and 6 nonresidues modulo 13, and there are 3 quadratic residues and 3 nonresidues modulo 7. Using our earlier observation that $(p - b)^2 \equiv b^2 \pmod{p}$, we can easily verify that there are an equal number of quadratic residues and nonresidues modulo any (odd) prime.

Theorem 23.1. *Let p be an odd prime. Then there are exactly $(p - 1)/2$ quadratic residues modulo p and exactly $(p - 1)/2$ nonresidues modulo p .*

Verification. The quadratic residues are the nonzero numbers that are squares modulo p , so they are the numbers

$$1^2, 2^2, \dots, (p - 1)^2 \pmod{p}.$$

But, as we noted above, we only need to go halfway,

$$1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2 \pmod{p},$$

since the same numbers are repeated in reverse order if we square the remaining numbers

$$\left(\frac{p+1}{2}\right)^2, \dots, (p-2)^2, (p-1)^2 \pmod{p}.$$

So in order to show that there are exactly $(p-1)/2$ quadratic residues, we need to check that the numbers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are all different modulo p .

Suppose that b_1 and b_2 are numbers between 1 and $(p-1)/2$, and suppose that $b_1^2 \equiv b_2^2 \pmod{p}$. We want to show that $b_1 = b_2$. The fact that $b_1^2 \equiv b_2^2 \pmod{p}$ means that

$$p \text{ divides } b_1^2 - b_2^2 = (b_1 - b_2)(b_1 + b_2).$$

However, $b_1 + b_2$ is between 2 and $p-1$, so it can't be divisible by p . Thus p must divide $b_1 - b_2$. But $|b_1 - b_2| < (p-1)/2$, so the only way for $b_1 - b_2$ to be divisible by p is to have $b_1 = b_2$. This shows that the numbers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are all different modulo p , so there are exactly $(p-1)/2$ quadratic residues modulo p . Now we need only observe that there are $p-1$ numbers between 1 and $p-1$, so if half of them are quadratic residues, the other half must be nonresidues. $\square$

Suppose that we take two quadratic residues and multiply them together. Do we get a QR or an NR, or do we sometimes get one and sometimes the other? For example, 3 and 10 are QRs modulo 13, and their product $3 \cdot 10 = 30 \equiv 4$ is again a QR modulo 13. Actually, this should have been clear without any computation, since if we multiply two squares, we should get a square. We can formally verify this in the following way. Suppose that a_1 and a_2 are both QRs modulo p . This means that there are numbers b_1 and b_2 so that $a_1 \equiv b_1^2 \pmod{p}$ and $a_2 \equiv b_2^2 \pmod{p}$. Multiplying these two congruences together, we find that $a_1 a_2 \equiv (b_1 b_2)^2 \pmod{p}$, which shows that $a_1 a_2$ is a QR.

The situation is less clear if we multiply a QR by an NR, or if we multiply two NRs together. Here are some examples using the data in our tables:

QR $\times$ NR $\equiv ?? \pmod{p}$	NR $\times$ NR $\equiv ?? \pmod{p}$
$2 \times 5 \equiv 3 \pmod{7}$ NR	$3 \times 5 \equiv 1 \pmod{7}$ QR
$5 \times 6 \equiv 8 \pmod{11}$ NR	$6 \times 7 \equiv 9 \pmod{11}$ QR
$4 \times 5 \equiv 7 \pmod{13}$ NR	$5 \times 11 \equiv 3 \pmod{13}$ QR
$10 \times 7 \equiv 5 \pmod{13}$ NR	$7 \times 11 \equiv 12 \pmod{13}$ QR

Thus, multiplying a quadratic residue and a nonresidue seems to yield a nonresidue, while the product of two nonresidues always seems to be a residue. Symbolically, we might write

$$\text{QR} \times \text{QR} = \text{QR}, \quad \text{QR} \times \text{NR} = \text{NR}, \quad \text{NR} \times \text{NR} = \text{QR}.$$

We've already seen that the first relation is true, and we now verify the other two relations.

Theorem 23.2 (Quadratic Residue Multiplication Rule). (Version 1) *Let p be an odd prime. Then:*

- (i) *The product of two quadratic residues modulo p is a quadratic residue.*
- (ii) *The product of a quadratic residue and a nonresidue is a nonresidue.*
- (iii) *The product of two nonresidues is a quadratic residue.*

These three rules can be summarized symbolically by the formulas

$$QR \times QR = QR, \quad QR \times NR = NR, \quad NR \times NR = QR.$$

Verification. We have already seen that $QR \times QR = QR$. Suppose next that a_1 is a QR, say $a_1 \equiv b_1^2 \pmod{p}$, and that a_2 is an NR. We are going to assume that $a_1 a_2$ is a QR and derive a contradiction. The assumption that $a_1 a_2$ is a QR means that it is congruent to b_3^2 for some b_3 , so we have

$$b_3^2 \equiv a_1 a_2 \equiv b_1^2 a_2 \pmod{p}.$$

Note that $\gcd(b_1, p) = 1$, since $p \nmid a_1$ and $a_1 = b_1^2$, so the Linear Congruence Theorem (Theorem 8.1) says that we can find an inverse for b_1 modulo p . In other words, we can find some c_1 such that $c_1 b_1 \equiv 1 \pmod{p}$. Multiplying both sides of the above congruence by c_1^2 gives

$$c_1^2 b_3^2 \equiv c_1^2 a_1 a_2 \equiv (c_1 b_1)^2 a_2 \equiv a_2 \pmod{p}.$$

Thus $a_2 \equiv (c_1 b_3)^2 \pmod{p}$ is a QR, contradicting the fact that a_2 is a NR. This completes the proof that

$$QR \times NR = NR.$$

We are left to deal with the product of two NRs. Let a be an NR and consider the set of values

$$a, 2a, 3a, \dots, (p-2)a, (p-1)a \pmod{p}.$$

By an argument we've used before (see Claim 9.2 on page 63), these are just the numbers $1, 2, \dots, (p-1)$ rearranged in some different order. In particular, they include the $\frac{1}{2}(p-1)$ QRs and the $\frac{1}{2}(p-1)$ NRs. However, as we already proved, each time that we multiply a by a QR, we get an NR, so the $\frac{1}{2}(p-1)$ products

$$a \times QR$$

already give us all $\frac{1}{2}(p-1)$ QRs in the list. Hence when we multiply a by an NR, the only possibility is that it is equal to one of the NRs in the list, because the $a \times$ QR products have already used up all of the QRs in the list.² $\square$

This completes the verification of the quadratic residue multiplication rules. Now take a minute to stare at

$$\text{QR} \times \text{QR} = \text{QR}, \quad \text{QR} \times \text{NR} = \text{NR}, \quad \text{NR} \times \text{NR} = \text{QR}.$$

Do these rules remind you of anything? If not, here's a hint. Suppose that we try to replace the symbols QR and NR with numbers. What numbers would work? That's right, the symbol QR behaves like $+1$ and the symbol NR behaves like -1 . Notice that the somewhat mysterious third rule, the one that says that the product of two nonresidues is a quadratic residue, reflects the equally mysterious rule³

$$(-1) \times (-1) = +1.$$

Having observed that QRs behave like $+1$ and NRs behave like -1 , Adrien-Marie Legendre introduced the following useful notation.

The *Legendre symbol* of a modulo p is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } a \text{ is a nonresidue modulo } p. \end{cases}$$

For example, data from our earlier tables says that

$$\left(\frac{3}{13}\right) = 1, \quad \left(\frac{11}{13}\right) = -1, \quad \left(\frac{2}{7}\right) = 1, \quad \left(\frac{3}{7}\right) = -1.$$

Using the Legendre symbol, our quadratic residue multiplication rules can be given by a single formula.

Theorem 23.3 (Quadratic Residue Multiplication Rule). (Version 2) *Let p be an odd prime. Then*

$$\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right).$$

²“When you have eliminated all of the quadratic residues, the remaining numbers, no matter how improbable, must be the nonresidues!” (with apologies to Sherlock Holmes and Sir Arthur Conan Doyle).

³You may no longer consider the formula $(-1) \times (-1) = +1$ mysterious, since it's so familiar to you. But you should have found it mysterious the first time you saw it. And if you stop to think about it, there is no obvious reason why the product of two negative numbers should equal a positive number. Can you come up with a convincing argument that $(-1) \times (-1)$ must equal $+1$?

The Legendre symbol is useful for making calculations. For example, suppose that we want to know if 75 is a square modulo 97. We can compute

$$\left(\frac{75}{97}\right) = \left(\frac{3 \cdot 5 \cdot 5}{97}\right) = \left(\frac{3}{97}\right) \left(\frac{5}{97}\right) \left(\frac{5}{97}\right) = \left(\frac{3}{97}\right).$$

Notice that it doesn't matter whether $\left(\frac{5}{97}\right)$ is $+1$ or -1 , since it appears twice, and $(+1)^2 = (-1)^2 = 1$. Now we observe that $10^2 \equiv 3 \pmod{97}$, so 3 is a QR. Hence,

$$\left(\frac{75}{97}\right) = \left(\frac{3}{97}\right) = 1.$$

Of course, we were lucky in being able to recognize 3 as a QR modulo 97. Is there some way to evaluate a Legendre symbol like $\left(\frac{3}{97}\right)$ without relying on luck or trial and error? The answer is yes, but that's a topic for another chapter.

Exercises

23.1. Make a list of all the quadratic residues and all the nonresidues modulo 19.

23.2.  Write a program that takes as input a prime p and produces as output the two numbers

$A = \text{sum of all } 1 \leq a < p \text{ such that } a \text{ is a quadratic residue modulo } p,$

$B = \text{sum of all } 1 \leq a < p \text{ such that } a \text{ is a nonresidue modulo } p.$

For example, if $p = 11$, then the quadratic residues are

$$\begin{aligned} 1^2 &\equiv 1 \pmod{11}, & 2^2 &\equiv 4 \pmod{11}, & 3^2 &\equiv 9 \pmod{11}, \\ 4^2 &\equiv 5 \pmod{11}, & 5^2 &\equiv 3 \pmod{11}, \end{aligned}$$

so

$$A = 1 + 4 + 9 + 5 + 3 = 22 \quad \text{and} \quad B = 2 + 6 + 7 + 8 + 10 = 33.$$

- Make a list of A and B for all primes $p < 100$.
- What is the value of $A + B$? Prove that your guess is correct.
- Compute $A \bmod p$ and $B \bmod p$. Find a pattern and prove that it is correct.
- For which primes is it true that $A = B$? After reading Chapter 24, prove that your guess is correct.
- If $A \neq B$, which one tends to be larger, A or B ? Try to prove that your guess is correct, but be forewarned that this is a very difficult problem.

23.3. A number a is called a *cubic residue modulo p* if it is congruent to a cube modulo p , that is, if there is a number b such that $a \equiv b^3 \pmod{p}$.

- (a) Make a list of all the cubic residues modulo 5, modulo 7, modulo 11, and modulo 13.
- (b) Find two numbers a_1 and b_1 such that neither a_1 nor b_1 is a cubic residue modulo 19, but a_1b_1 is a cubic residue modulo 19. Similarly, find two numbers a_2 and b_2 such that none of the three numbers a_2 , b_2 , or a_2b_2 is a cubic residue modulo 19.
- (c) If $p \equiv 2 \pmod{3}$, make a conjecture as to which a 's are cubic residues. Prove that your conjecture is correct.

Chapter 24

Is -1 a Square Modulo p ? Is 2?

Revised Version of Chapter 24

In the previous chapter we took various primes p and looked at the a 's that were quadratic residues and the a 's that were nonresidues. For example, we made a table of squares modulo 13 and used the table to see that 3 and 12 are QRs modulo 13, while 2 and 5 are NRs modulo 13.

In keeping with all of the best traditions of mathematics, we now turn this problem on its head. Rather than taking a particular prime p and listing the a 's that are QRs and NRs, we instead fix an a and ask for which primes p is a a QR. To make it clear exactly what we're asking, we start with the particular value $a = -1$. The question that we want to answer is as follows:

For which primes p is -1 a QR?

We can rephrase this question in other ways, such as “For which primes p does the congruence $x^2 \equiv -1 \pmod{p}$ have a solution?” and “For which primes p is $\left(\frac{-1}{p}\right) = 1$?”

As always, we need some data before we can make any hypotheses. We can answer our question for small primes in the usual mindless way by making a table of $1^2, 2^2, 3^2, \dots \pmod{p}$ and checking if any of the numbers are congruent to -1 modulo p . So, for example, -1 is not a square modulo 3, since $1^2 \not\equiv -1 \pmod{3}$ and $2^2 \not\equiv -1 \pmod{3}$, while -1 is a square modulo 5, since $2^2 \equiv -1 \pmod{5}$. Here's a more extensive list.

p	3	5	7	11	13	17	19	23	29	31
Solution(s) to $x^2 \equiv -1 \pmod{p}$	NR	2, 3	NR	NR	5, 8	4, 13	NR	NR	12, 17	NR

Reading from this table, we compile the following data:

-1 is a quadratic residue for $p = 5, 13, 17, 29$.

-1 is a nonresidue for $p = 3, 7, 11, 19, 23, 31$.

It's not hard to discern the pattern. If p is congruent to 1 modulo 4, then -1 seems to be a quadratic residue modulo p , and if p is congruent to 3 modulo 4, then -1 seems to be a nonresidue. We can express this guess using Legendre symbols,

$$\left(\frac{-1}{p}\right) \stackrel{?}{=} \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Let's check our conjecture on the next few cases. The next two primes, 37 and 41, are both congruent to 1 modulo 4 and, sure enough,

$x^2 \equiv -1 \pmod{37}$ has the solutions $x \equiv 6$ and $31 \pmod{37}$, and

$x^2 \equiv -1 \pmod{41}$ has the solutions $x \equiv 9$ and $32 \pmod{41}$.

Similarly, the next two primes 43 and 47 are congruent to 3 modulo 4, and we check that -1 is a nonresidue for 43 and 47. Our guess is looking good!

The tool that we use to verify our conjecture might be called the "Square Root of Fermat's Little Theorem." How, you may well ask, does one take the square root of a theorem? Recall that Fermat's Little Theorem (Chapter 9) says

$$a^{p-1} \equiv 1 \pmod{p}.$$

We won't really be taking the square root of this theorem, of course. Instead, we take the square root of the quantity a^{p-1} and ask for its value. So we want to answer the following question:

Let $A = a^{(p-1)/2}$. What is the value of A modulo p ?

One thing is obvious. If we square A , then Fermat's Little Theorem tells us that

$$A^2 = a^{p-1} \equiv 1 \pmod{p}.$$

Hence, p divides $A^2 - 1 = (A - 1)(A + 1)$, so either p divides $A - 1$ or p divides $A + 1$. (Notice how we are using the property of prime numbers proved on page 44.) Thus A must be congruent to either $+1$ or -1 .

Here are a few random values of p , a , and A . For comparison purposes, we have also included the value of the Legendre symbol $\left(\frac{a}{p}\right)$. Do you see a pattern?

p	11	31	47	97	173	409	499	601	941	1223
a	3	7	10	15	33	78	33	57	222	129
$A \pmod{p}$	1	1	-1	-1	1	-1	1	-1	1	1
$\left(\frac{a}{p}\right)$	1	1	-1	-1	1	-1	1	-1	1	1

It certainly appears that $A \equiv 1 \pmod{p}$ when a is a quadratic residue and that $A \equiv -1 \pmod{p}$ when a is a nonresidue. In other words, it looks like $A \pmod{p}$ has the same value as the Legendre symbol $\left(\frac{a}{p}\right)$. We use a counting argument to verify this assertion, which goes by the name of Euler's Criterion.

Theorem 24.1 (Euler's Criterion). *Let p be an odd prime. Then*

$$a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}.$$

Verification. Suppose first that a is a quadratic residue, say $a \equiv b^2 \pmod{p}$. Then Fermat's Little Theorem (Theorem 9.1) tells us that

$$a^{(p-1)/2} \equiv (b^2)^{(p-1)/2} = b^{p-1} \equiv 1 \pmod{p}.$$

Hence $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}$, which is Euler's criterion when a is a quadratic residue.

We next consider the congruence

$$X^{(p-1)/2} - 1 \equiv 0 \pmod{p}.$$

We have just proven that every quadratic residue is a solution to this congruence, and we know from Theorem 23.1 that there are exactly $\frac{1}{2}(p-1)$ distinct quadratic residues. We also know from the Polynomial Roots Mod p Theorem (Theorem 8.2) that this polynomial congruence can have at most $\frac{1}{2}(p-1)$ distinct solutions. Hence

$$\{\text{solutions to } X^{(p-1)/2} - 1 \equiv 0 \pmod{p}\} = \{\text{quadratic residues modulo } p\}.$$

Now let a be a nonresidue. Fermat's Little Theorem tells us that $a^{p-1} \equiv 1 \pmod{p}$, so

$$0 \equiv a^{p-1} - 1 \equiv (a^{(p-1)/2} - 1)(a^{(p-1)/2} + 1) \pmod{p}.$$

The first factor is not zero modulo p , because we already showed that the solutions to $X^{(p-1)/2} - 1 \equiv 0 \pmod{p}$ are the quadratic residues. Hence the second factor must vanish modulo p , so

$$a^{(p-1)/2} \equiv -1 = \left(\frac{a}{p}\right) \pmod{p}.$$

This shows that Euler's criterion is also true for nonresidues. $\square$

Using Euler's Criterion, it is very easy to determine if -1 is a quadratic residue modulo p . For example, if we want to know whether -1 is a square modulo the prime $p = 6911$, we just need to compute

$$(-1)^{(6911-1)/2} = (-1)^{3455} = -1.$$

Euler's Criterion then tells us that

$$\left(\frac{-1}{6911}\right) \equiv -1 \pmod{6911}.$$

But $\left(\frac{a}{p}\right)$ is always either $+1$ or -1 , so in this case we must have $\left(\frac{-1}{6911}\right) = -1$. Hence, -1 is a nonresidue modulo 6911 .

Similarly, for the prime $p = 7817$ we find that

$$(-1)^{(7817-1)/2} = (-1)^{3908} = 1.$$

Hence, $\left(\frac{-1}{7817}\right) = 1$, so -1 is a quadratic residue modulo 7817 . Observe that, although we now know that the congruence

$$x^2 \equiv -1 \pmod{7817}$$

is solvable, we still don't have any efficient way to find a solution. The solutions turn out to be $x \equiv 2564 \pmod{7817}$ and $x \equiv 5253 \pmod{7817}$.

As these two examples make clear, Euler's Criterion can be used to determine exactly which primes have -1 as a quadratic residue. This elegant result, which answers the initial question in the title of this chapter, is the first part of the Law of Quadratic Reciprocity.

Theorem 24.2 (Quadratic Reciprocity). (Part I) *Let p be an odd prime. Then*

$$\begin{aligned} -1 \text{ is a quadratic residue modulo } p & \quad \text{if } p \equiv 1 \pmod{4}, \text{ and} \\ -1 \text{ is a nonresidue modulo } p & \quad \text{if } p \equiv 3 \pmod{4}. \end{aligned}$$

In other words, using the Legendre symbol,

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

Verification. Euler's Criterion says that

$$(-1)^{(p-1)/2} \equiv \left(\frac{-1}{p}\right) \pmod{p}.$$

Suppose first that $p \equiv 1 \pmod{4}$, say $p = 4k + 1$. Then

$$(-1)^{(p-1)/2} = (-1)^{2k} = 1, \quad \text{so} \quad 1 \equiv \left(\frac{-1}{p}\right) \pmod{p}.$$

But $\left(\frac{-1}{p}\right)$ is either $+1$ or -1 , so it must equal 1 . This proves that if $p \equiv 1 \pmod{4}$ then $\left(\frac{-1}{p}\right) = 1$.

Next we suppose that $p \equiv 3 \pmod{4}$, say $p = 4k + 3$. Then

$$(-1)^{(p-1)/2} = (-1)^{2k+1} = -1, \quad \text{so} \quad -1 \equiv \left(\frac{-1}{p}\right) \pmod{p}.$$

This shows that $\left(\frac{-1}{p}\right)$ must equal -1 , which completes the verification of Quadratic Reciprocity (Part I). $\square$

We can use the first part of quadratic reciprocity to answer a question left over from Chapter 12. As you may recall, we showed that there are infinitely many primes that are congruent to 3 modulo 4 , but we left unanswered the analogous question for primes congruent to 1 modulo 4 .

Theorem 24.3 (Primes 1 (Mod 4) Theorem). *There are infinitely many primes that are congruent to 1 modulo 4.*

Proof. Suppose we are given a list of primes $p_1, p_2, \dots, p_r$, all of which are congruent to 1 modulo 4 . We are going to find a new prime, not in our list, that is congruent to 1 modulo 4 . Repeating this process gives a list of any desired length.

Consider the number

$$A = (2p_1p_2 \cdots p_r)^2 + 1.$$

We know that A can be factored into a product of primes, say

$$A = q_1q_2 \cdots q_s.$$

It is clear that $q_1, q_2, \dots, q_s$ are not in our original list, since none of the p_i 's divide A . So all we need to do is show that at least one of the q_i 's is congruent to 1 modulo 4 . In fact, we'll see that all of them are.

First we note that A is odd, so all the q_i 's are odd. Next, each q_i divides A , so

$$(2p_1p_2 \cdots p_r)^2 + 1 = A \equiv 0 \pmod{q_i}.$$

This means that $x = 2p_1p_2 \cdots p_r$ is a solution to the congruence

$$x^2 \equiv -1 \pmod{q_i},$$

so -1 is a quadratic residue modulo q_i . Now Quadratic Reciprocity tells us that $q_i \equiv 1 \pmod{4}$. $\square$

We can use the procedure described in this proof to produce a list of primes that are congruent to 1 modulo 4. Thus, if we start with $p_1 = 5$, then we form $A = (2p_1)^2 + 1 = 101$, so our second prime is $p_2 = 101$. Then

$$A = (2p_1p_2)^2 + 1 = 1020101,$$

which is again prime, so our third prime is $p_3 = 1020101$. We'll go one more step,

$$\begin{aligned} A &= (2p_1p_2p_3)^2 + 1 \\ &= 1061522231810040101 \\ &= 53 \cdot 1613 \cdot 12417062216309. \end{aligned}$$

Notice that all the primes 53, 1613, and 12417062216309 are congruent to 1 modulo 4, just as predicted by the theory.

Having successfully answered the first question in the title of this chapter, we move on to the second question and consider $a = 2$, that "oddest" of all primes. Just as we did with $a = -1$, we are looking for some simple characterization for the primes p such that 2 is a quadratic residue modulo p . Can you find the pattern in the following data, where the line labeled $x^2 \equiv 2$ gives the solutions to $x^2 \equiv 2 \pmod{p}$ if 2 is a quadratic residue modulo p and is marked NR if 2 is a nonresidue?

p	3	5	7	11	13	17	19	23	29	31
$x^2 \equiv 2$	NR	NR	3, 4	NR	NR	6, 11	NR	5, 18	NR	8, 23

p	37	41	43	47	53	59	61	67	71	73
$x^2 \equiv 2$	NR	17, 24	NR	7, 40	NR	NR	NR	NR	12, 59	32, 41

p	79	83	89	97	101	103	107	109	113	127
$x^2 \equiv 2$	9, 70	NR	25, 64	14, 83	NR	38, 65	NR	NR	51, 62	16, 111

Here's the list of primes separated according to whether 2 is a residue or a non-residue.

$$\begin{aligned} 2 \text{ is a quadratic residue for } p &= 7, 17, 23, 31, 41, 47, 71, 73, \\ &\quad 79, 89, 97, 103, 113, 127 \\ 2 \text{ is a nonresidue for } p &= 3, 5, 11, 13, 19, 29, 37, 43, 53, 59, \\ &\quad 61, 67, 83, 101, 107, 109 \end{aligned}$$

For $a = -1$, it turned out that the congruence class of p modulo 4 was crucial. Is there a similar pattern if we reduce these two lists of primes modulo 4? Here's what happens if we do.

$$\begin{aligned} 7, 17, 23, 31, 41, 47, 71, 73, 79, 89, 97, 103, 113, 127 \\ \equiv 3, 1, 3, 3, 1, 3, 3, 1, 3, 1, 1, 3, 1, 3 \pmod{4}, \\ 3, 5, 11, 13, 19, 29, 37, 43, 53, 59, 61, 67, 83, 101, 107, 109 \\ \equiv 3, 1, 3, 1, 3, 1, 1, 3, 1, 3, 1, 3, 3, 1, 3, 1 \pmod{4}. \end{aligned}$$

This doesn't look too promising. Maybe we should try reducing modulo 3.

$$\begin{aligned} 7, 17, 23, 31, 41, 47, 71, 73, 79, 89, 97, 103, 113, 127 \\ \equiv 1, 2, 2, 1, 2, 2, 2, 1, 1, 2, 1, 1, 2, 1 \pmod{3} \\ 3, 5, 11, 13, 19, 29, 37, 43, 53, 59, 61, 67, 83, 101, 107, 109 \\ \equiv 0, 2, 2, 1, 1, 2, 1, 1, 2, 2, 1, 1, 2, 2, 2, 1 \pmod{3}. \end{aligned}$$

This doesn't look any better. Let's make one more attempt before we give up. What happens if we reduce modulo 8?

$$\begin{aligned} 7, 17, 23, 31, 41, 47, 71, 73, 79, 89, 97, 103, 113, 127 \\ \equiv 7, 1, 7, 7, 1, 7, 7, 1, 7, 1, 1, 7, 1, 7 \pmod{8} \\ 3, 5, 11, 13, 19, 29, 37, 43, 53, 59, 61, 67, 83, 101, 107, 109 \\ \equiv 3, 5, 3, 5, 3, 5, 5, 3, 5, 3, 3, 5, 3, 5 \pmod{8}. \end{aligned}$$

Eureka! It surely can't be a coincidence that the first line is all 1's and 7's and the second line is all 3's and 5's. This suggests the general rule that 2 is a quadratic residue modulo p if p is congruent to 1 or 7 modulo 8 and that 2 is a nonresidue if p is congruent to 3 or 5 modulo 8. In terms of Legendre symbols, we would write

$$\left(\frac{2}{p}\right) \stackrel{?}{=} \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8}, \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8}. \end{cases}$$

Can we use Euler's Criterion to verify our guess? Unfortunately, the answer is no, or at least not in any obvious way, since there doesn't seem to be an easy method to calculate $2^{(p-1)/2} \pmod{p}$. However, if you go back and examine our proof of Fermat's Little Theorem in Chapter 9, you'll see that we took the numbers $1, 2, \dots, p-1$, multiplied each one by a , and then multiplied them all together. This gave us a factor of a^{p-1} to pull out. In order to use Euler's Criterion, we only want $\frac{1}{2}(p-1)$ factors of a to pull out, so rather than starting with all the numbers from 1 to p , we just take the numbers from 1 to $\frac{1}{2}(p-1)$. We illustrate this idea, which is due to Gauss, to determine if 2 is a quadratic residue modulo 13.

We begin with half the numbers from 1 to 12: 1, 2, 3, 4, 5, 6. If we multiply each by 2 and then multiply them together, we get

$$\begin{aligned} 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 &= (2 \cdot 1)(2 \cdot 2)(2 \cdot 3)(2 \cdot 4)(2 \cdot 5)(2 \cdot 6) \\ &= 2^6 \cdot 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \\ &= 2^6 \cdot 6!. \end{aligned}$$

Notice the factor of $2^6 = 2^{(13-1)/2}$, which is the number we're really interested in.

Gauss's idea is to take the numbers 2, 4, 6, 8, 10, 12 and reduce each of them modulo 13 to get a number lying between -6 and 6. The first three stay the same, but we need to subtract 13 from the last three to get them into this range. Thus,

$$\begin{array}{lll} 2 \equiv 2 \pmod{13} & 4 \equiv 4 \pmod{13} & 6 \equiv 6 \pmod{13} \\ 8 \equiv -5 \pmod{13} & 10 \equiv -3 \pmod{13} & 12 \equiv -1 \pmod{13}. \end{array}$$

Multiplying these numbers together, we find that

$$\begin{aligned} 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 &\equiv 2 \cdot 4 \cdot 6 \cdot (-5) \cdot (-3) \cdot (-1) \\ &\equiv (-1)^3 \cdot 2 \cdot 4 \cdot 6 \cdot 5 \cdot 3 \cdot 1 \\ &\equiv -6! \pmod{13}. \end{aligned}$$

Equating these two values of $2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \pmod{13}$, we see that

$$2^6 \cdot 6! \equiv -6! \pmod{13}.$$

This implies that $2^6 \equiv -1 \pmod{13}$, so Euler's Criterion tells us that 2 is a non-residue modulo 13.

Let's briefly use the same ideas to check if 2 is a quadratic residue modulo 17. We take the numbers from 1 to 8, multiply each by 2, multiply them together, and calculate the product in two different ways. The first way gives

$$2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 16 = 2^8 \cdot 8!.$$

For the second way, we reduce modulo 17 to bring the numbers into the range from -8 to 8. Thus,

$$\begin{array}{lll} 2 \equiv 2 \pmod{17} & 4 \equiv 4 \pmod{17} & 6 \equiv 6 \pmod{17} \\ 8 \equiv 8 \pmod{17} & 10 \equiv -7 \pmod{17} & 12 \equiv -5 \pmod{17} \\ 14 \equiv -3 \pmod{17} & 16 \equiv -1 \pmod{17}. \end{array}$$

Multiplying these together gives

$$\begin{aligned} 2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdot 14 \cdot 16 &\equiv 2 \cdot 4 \cdot 6 \cdot 8 \cdot (-7) \cdot (-5) \cdot (-3) \cdot (-1) \\ &\equiv (-1)^4 \cdot 8! \pmod{17}. \end{aligned}$$

Therefore, $2^8 \cdot 8! \equiv (-1)^4 \cdot 8! \pmod{17}$, so $2^8 \equiv 1 \pmod{17}$, and hence 2 is a quadratic residue modulo 17.

Now let's think about Gauss's method a little more generally. Let p be any odd prime. To make our formulas simpler, we let

$$P = \frac{p-1}{2}.$$

We start with the even numbers $2, 4, 6, \dots, p-1$. Multiplying them together and factoring out a 2 from each number gives

$$2 \cdot 4 \cdot 6 \cdots (p-1) = 2^{(p-1)/2} \cdot 1 \cdot 2 \cdot 3 \cdots \frac{p-1}{2} = 2^P \cdot P!.$$

The next step is to take the list $2, 4, 6, \dots, p-1$ and reduce each number modulo p so that it lies in the range from $-P$ to P , that is, between $-(p-1)/2$ and $(p-1)/2$. The first few numbers won't change, but at some point in the list we'll start hitting numbers that are larger than $(p-1)/2$, and each of these large numbers needs to have p subtracted from it. Notice that the number of minus signs introduced is exactly the number of times we need to subtract p . In other words,

$$\text{Number of minus signs} = \binom{\text{Number of integers in the list}}{\substack{2, 4, 6, \dots, (p-1) \\ \text{that are larger than } \frac{1}{2}(p-1)}}.$$

The following illustration may help to explain this procedure.

$$\underbrace{2 \cdot 4 \cdot 6 \cdot 8 \cdot 10 \cdot 12 \cdots}_{\substack{\text{Numbers } \leq (p-1)/2 \\ \text{are left unchanged.}}} \mid \underbrace{\cdots (p-5) \cdot (p-3) \cdot (p-1)}_{\substack{\text{Numbers } > (p-1)/2. \\ \text{Need to subtract } p \text{ from each.}}}$$

Comparing the two products, we get

$$2^P \cdot P! = 2 \cdot 4 \cdot 6 \cdots (p-1) \equiv (-1)^{(\text{Number of minus signs})} \cdot P! \pmod{p},$$

so canceling $P!$ from each side gives the fundamental formula

$$2^{(p-1)/2} \equiv (-1)^{(\text{Number of minus signs})} \pmod{p}.$$

Using this formula, it is easy to verify our earlier guess, thereby answering the second question in the chapter title.

Theorem 24.4 (Quadratic Reciprocity). (Part II) *Let p be an odd prime. Then 2 is a quadratic residue modulo p if p is congruent to 1 or 7 modulo 8, and 2 is a nonresidue modulo p if p is congruent to 3 or 5 modulo 8. In terms of the Legendre symbol,*

$$\left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \text{ or } 7 \pmod{8}, \\ -1 & \text{if } p \equiv 3 \text{ or } 5 \pmod{8}. \end{cases}$$

Verification. There are actually four cases to consider, depending on the value of $p \pmod{8}$. We do two of them and leave the other two for you.

We start with the case that $p \equiv 3 \pmod{8}$, say $p = 8k + 3$. We need to list the numbers $2, 4, \dots, p-1$ and determine how many of them are larger than $\frac{1}{2}(p-1)$. In this case, $p-1 = 8k+2$ and $\frac{1}{2}(p-1) = 4k+1$, so the cutoff is as indicated in the following diagram:

$$2 \cdot 4 \cdot 6 \cdots 4k \quad \Bigg| \quad (4k+2) \cdot (4k+4) \cdots (8k+2).$$

We need to count how many numbers there are to the right of the vertical bar. In other words, how many even numbers are there between $4k+2$ and $8k+2$? The answer is $2k+1$. (If this isn't clear to you, try a few values for k and you'll see why it's correct.) This shows that there are $2k+1$ minus signs, so the fundamental formula given above tells us that

$$2^{(p-1)/2} \equiv (-1)^{2k+1} \equiv -1 \pmod{p}.$$

Now Euler's Criterion says that 2 is a nonresidue, so we have proved that 2 is a nonresidue for any prime p that is congruent to 3 modulo 8.

Next let's look at the primes that are congruent to 7 modulo 8, say $p = 8k + 7$. Now the even numbers $2, 4, \dots, p-1$ are the numbers from 2 to $8k+6$, and the midpoint is $\frac{1}{2}(p-1) = 4k+3$. The cutoff in this case is

$$2 \cdot 4 \cdot 6 \cdots (4k+2) \quad \Bigg| \quad (4k+4) \cdot (4k+6) \cdots (8k+6).$$

There are exactly $2k+2$ numbers to the right of the vertical bar, so we get $2k+2$ minus signs. This yields

$$2^{(p-1)/2} \equiv (-1)^{2k+2} \equiv 1 \pmod{p},$$

so Euler's criterion tells us that 2 is a quadratic residue. This proves that 2 is a quadratic residue for any prime p that is congruent to 7 modulo 8. $\square$

Exercises

24.1. Determine whether each of the following congruences has a solution. (All of the moduli are primes.)

- (a) $x^2 \equiv -1 \pmod{5987}$ (c) $x^2 + 14x - 35 \equiv 0 \pmod{337}$
 (b) $x^2 \equiv 6780 \pmod{6781}$ (d) $x^2 - 64x + 943 \equiv 0 \pmod{3011}$

[Hint. For (c), use the quadratic formula to find out what number you need to take the square root of modulo 337, and similarly for (d).]

24.2. Use the procedure described in the Primes 1 (Mod 4) Theorem to generate a list of primes congruent to 1 modulo 4, starting with the seed $p_1 = 17$.

24.3. Here is a list of the first few primes for which 3 is a quadratic residue and a non-residue.

Quadratic Residue: $p = 11, 13, 23, 37, 47, 59, 61, 71, 73, 83, 97, 107, 109$

Nonresidue: $p = 5, 7, 17, 19, 29, 31, 41, 43, 53, 67, 79, 89, 101, 103, 113, 127$

Try reducing this list modulo m for various m 's until you find a pattern, and make a conjecture explaining which primes have 3 as a quadratic residue.

24.4. Finish the verification of Quadratic Reciprocity Part II for the other two cases: primes congruent to 1 modulo 8 and primes congruent to 5 modulo 8.

24.5. Use the same ideas we used to verify Quadratic Reciprocity (Part II) to verify the following two assertions.

- (a) If p is congruent to 1 modulo 5, then 5 is a quadratic residue modulo p .
 (b) If p is congruent to 2 modulo 5, then 5 is a nonresidue modulo p .

[Hint. Reduce the numbers $5, 10, 15, \dots, \frac{5}{2}(p-1)$ so that they lie in the range from $-\frac{1}{2}(p-1)$ to $\frac{1}{2}(p-1)$ and check how many of them are negative.]

24.6. Suppose that q is a prime number that is congruent to 1 modulo 4, and suppose that the number $p = 2q + 1$ is also a prime number. (For example, q could equal 5 and p equal 11.) Show that 2 is a primitive root modulo p .